SQL Analysis

```sql
# 1. Total Sales
SELECT SUM(Purchase) AS Total_Sales
FROM black_friday_data;
```

100% 24:3

Result Grid Filter Rows: Search Export:

Total_Sales
5095812742

Explanation:

This query calculates the total sales generated from all customer purchases in the Black Friday dataset.

```sql
# 2. Average Purchase
SELECT ROUND(AVG(Purchase),2) AS Average_Purchase
FROM black_friday_data;
```

100% 24:7

Result Grid Filter Rows: Search Export:

Average_Purchase
9263.97

Explanation:

This query calculates the average purchase amount per transaction, helping to understand overall customer spending behaviour.

```sql
# 3. Maximum and Minimum Purchase
SELECT
MAX(Purchase) AS Highest_Purchase,
MIN(Purchase) AS Lowest_Purchase
FROM black_friday_data;
```

100% 24:13

Result Grid Filter Rows: Search Export:

Highest_Purcha...	Lowest_Purchase
23961	12

Explanation:

This query identifies the highest and lowest purchase amounts recorded in the dataset.

```sql
SELECT
Gender,
COUNT(*) AS Total_Orders,
SUM(Purchase) AS Total_Sales,
ROUND(AVG(Purchase),2) AS Avg_Purchase
FROM black_friday_data
GROUP BY Gender;
```

100% 17:22

Result Grid Filter Rows: Search Export:

Gender	Total_Orders	Total_Sales	Avg_Purchase
F	135809	1186232642	8734.57
M	414259	3909580100	9437.53

Explanation:

This query groups the data by gender and calculates the total number of orders, total sales, and average purchase amount for male and female customers. It helps compare purchasing behavior between genders

```sql
# 5. Age Group Analysis
SELECT
Age,
COUNT(*) AS Customers,
SUM(Purchase) AS Total_Sales,
ROUND(AVG(Purchase),2) AS Avg_Purchase
FROM black_friday_data
GROUP BY Age
ORDER BY Avg_Purchase DESC;
```

Result Grid | Filter Rows: Search | Export:

Gender	Total_Orders	Total_Sales	Avg_Purchase
F	135809	1186232642	8734.57
M	414259	3909580100	9437.53

Age Group Analysis

This query analyzes customer purchases across different age groups by calculating total orders, total sales, and average purchase amount.

```sql
# 6. City Category Analysis
SELECT
City_Category,
COUNT(*) AS Customers,
SUM(Purchase) AS Total_Sales
FROM black_friday_data
GROUP BY City_Category;
```

Result Grid | Filter Rows: Search | Export:

City_Category	Customers	Total_Sales
A	147720	1316471661
C	171175	1663807476
B	231173	2115533605

City Category Analysis

This query compares sales performance across different city categories, helping identify which locations generate the highest revenue.

```sql
# 7. Top 10 Selling Products
SELECT
Product_ID,
COUNT(*) AS Times_Sold,
SUM(Purchase) AS Revenue
FROM black_friday_data
GROUP BY Product_ID
ORDER BY Revenue DESC
LIMIT 10;
```

Result Grid | Filter Rows: Search | Export: | Fetch rows:

Product_ID	Times_Sold	Revenue
P00025442	1615	27995166
P00110742	1612	26722309
P00255842	1383	25168963
P00059442	1406	24338343
P00184942	1440	24334887
P00112142	1562	24216006
P00110942	1360	23639564
P00237542	1394	23425576
P00057642	1470	23102780
P00010742	1350	22164153

Explanation:

This query identifies the top 10 best-selling products based on total sales (or total purchase amount). It helps determine which products contribute the most to overall revenue and customer demand..

```sql
# 8. Most Popular Product Categories
SELECT
Product_Category_1,
COUNT(*) AS Orders,
SUM(Purchase) AS Revenue
FROM black_friday_data
GROUP BY Product_Category_1
ORDER BY Revenue DESC;
```

100% 23:59

Result Grid | Filter Rows: Search | Export: | Fetch rows:

Product_ID	Times_Sold	Revenue
P00025442	1615	27995166
P00110742	1612	26722309
P00255842	1383	25168963
P00059442	1406	24338343
P00184942	1440	24334887
P00112142	1562	24216006
P00110942	1360	23639564
P00237542	1394	23425576
P00057642	1470	23102780
P00010742	1350	22164153

Product Category Analysis

This query analyses sales for different product categories to determine which categories are the most popular and generate the highest revenue.

```sql
# 9. Married vs Unmarried Spending
SELECT
Marital_Status,
COUNT(*) AS Orders,
SUM(Purchase) AS Total_Sales,
ROUND(AVG(Purchase),2) AS Avg_Purchase
FROM black_friday_data
GROUP BY Marital_Status;
```

100% 25:68

Result Grid | Filter Rows: Search | Export:

Marital_Stat...	Orders	Total_Sales	Avg_Purchase
0	324731	3008927447	9265.91
1	225337	2086885295	9261.17

Explanation:

This query compares married and unmarried customers by calculating the total number of orders, total sales, and average purchase amount. It helps identify differences in purchasing behaviour based on marital status.

```sql
# 10. Occupation-wise Spending
SELECT
Occupation,
COUNT(*) AS Orders,
SUM(Purchase) AS Total_Sales
FROM black_friday_data
GROUP BY Occupation
ORDER BY Total_Sales DESC;
```

100% 25:68

Result Grid | Filter Rows: Search | Export:

Occupation	Orders	Total_Sales
4	72308	666244484
0	69638	635406958
7	59133	557371587
1	47426	424614144
17	40043	393281453
12	31179	305449446
20	33562	296570442
14	27309	259454692
16	25371	238346955
2	26588	238028583
6	20355	188416784
3	17650	162002168
15	12165	118960211
10	12930	115844465
5	12177	113649759
11	11586	106751618
19	8461	73700617
13	7728	71919481
18	6622	60721461
9	6291	54340046
8	1546	14737388

Occupation Analysis

This query analyzes customer spending based on occupation by calculating total orders, total sales, and average purchase amount.